

Closing the Reliability Gap: Causal Calibration of Saliency Maps via Perturbation-Aware Temperature Scaling and Significance Bucketing

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Abstract

The insertion/deletion tests of Petsiuk et al. (2018) are the de-facto causal yardstick for saliency maps: pixels are ranked by attributed importance and progressively removed or revealed while the model’s confidence is tracked. We study a *reliability gap* in this protocol: the perturbed images that the test feeds to the network are heavily out-of-distribution (OOD), so the softmax confidence that the curve is built from is itself untrustworthy, biasing the area-under-curve (AUC) score and producing noisy, non-monotone curves. Following the ReCalX framework (ReCalX, 2025), we recalibrate each model with *perturbation-aware temperature scaling*: one temperature per perturbation bin, fitted by negative-log-likelihood minimisation, applied without changing the model’s predictions. We further propose *significance bucketing*, a rank-based perturbation schedule that removes equal-count groups of pixels instead of a fixed top- $n\%$, making the test scale-invariant and robust to sparse saliency maps. Across four ImageNet architectures (ResNet50/101, ViT-B/16, ViT-B/32) and eight attribution methods, calibration lowers expected calibration error (ECE $0.28 \rightarrow 0.15$) and thresholded adaptive calibration error (TACE $0.35 \rightarrow 0.16$) on ResNet50, significantly increases deletion AUC, and makes the causal curves significantly more monotone (Spearman r : $-0.82 \rightarrow -0.94$, Wilcoxon $p < 0.01$), with the largest gains on perturbation-based attributions. Significance bucketing is in turn significantly more monotone than fixed-step removal ($p < 0.001$). Together the two ingredients make insertion/deletion a more trustworthy XAI metric.

1. Introduction

Attribution (saliency) methods explain an image classifier by assigning each input pixel a score for how much it supports a target class. To decide whether such an explanation is *faithful*, the community relies on the causal insertion/deletion test (Petsiuk et al., 2018): rank pixels by importance, then delete (or insert) them most-important-first and watch the predicted probability. A faithful map makes confidence collapse quickly under deletion (low AUC) and recover quickly under insertion (high AUC).

This report investigates a subtle but consequential flaw in that recipe and how to fix it. Our three highlights are:

- **The reliability gap is real and measurable.** Masking pixels pushes inputs off the data manifold; the classifier becomes miscalibrated (ECE up to 0.35 at high perturbation), so the confidence curve underlying the AUC is not trustworthy.
- **Perturbation-aware temperature scaling closes much of the gap.** A single temperature per perturbation level, fitted post-hoc and leaving predictions unchanged, cuts ECE/TACE roughly in half and makes deletion curves significantly more monotone.
- **Rank-based “significance bucketing” beats fixed-step removal.** Removing equal-count importance buckets is scale-invariant, robust to sparse maps (IG, Saliency), and yields significantly more monotone curves than the classic top- $n\%$ schedule.

Research questions. (RQ1) Does perturbation-aware recalibration restore reliable confidence under insertion/deletion perturbations, and does this make the resulting causal curves more stable (monotone) and their AUC less biased? **(RQ2)** Does significance bucketing provide a more sensitive and monotone causal evaluation than fixed-step top- $n\%$ removal, especially for skewed saliency maps?

Contributions. (C1) A reproducible pipeline integrating eight attribution methods, perturbation-binned temperature scaling (ReCalX), and two families of perturbation generators (fixed-step and rank-based bucketing) across four ImageNet architectures. **(C2)** An empirical study quantifying the reliability gap and showing that recalibration and bucketing each independently and significantly improve the trustworthiness of insertion/deletion evaluation. This project is directly related to the XAI course: it concerns the evaluation and improvement of post-hoc explanations for deep vision models.

2. Methodology

2.1 Attribution methods: inputs, outputs, meaning

An attribution method takes a trained model f , an input image $x \in \mathbb{R}^{3 \times 224 \times 224}$ and a target class c , and returns a 2D importance map $S \in \mathbb{R}^{224 \times 224}$ (channels summed) where S_{ij} scores how much pixel (i, j) supports $f_c(x)$. We use eight methods spanning two families. *Gradient/back-propagation* methods — Saliency (Simonyan et al., 2014), Integrated Gradients (IG) (Sundararajan et al., 2017) and GradientSHAP (Lundberg and Lee, 2017) — read importance from gradients of the class score. *Perturbation* methods — Occlusion (Zeiler and Fergus, 2014), LIME (Ribeiro et al., 2016), Feature Ablation / Shapley Value Sampling, and RISE (Petsiuk et al., 2018) — probe f with masked inputs and infer importance from the output change.

RISE is our default explainer because it is black-box and aligns naturally with the perturbation philosophy of the causal test. It samples N low-resolution binary masks ($s \times s$ grid, on-probability p_1), bilinearly upsamples and randomly shifts them to image size, multiplies each with x , queries the model, and estimates importance as the probability-weighted average of the masks: $S = \frac{1}{N p_1} \sum_m f_c(x \odot m) m$. We use $N=5000$, $s=10$, $p_1=0.1$ for ResNet50 and $N=2000$ for the other architectures (speed).

2.2 The causal insertion/deletion pipeline

Given S , pixels are sorted by importance. *Deletion* replaces the top-ranked pixels with a baseline (black) in steps and records f_c ; a faithful map yields a fast drop and low AUC. *Insertion* starts from a Gaussian-blur baseline and restores top pixels; a faithful map yields a fast rise and high AUC. We compute the AUC of the confidence-vs-fraction curve (`sklearn.metrics.auc`, trapezoidal), and two stability statistics: the *Spearman* rank correlation between perturbation level and confidence (ideal -1 for deletion, $+1$ for insertion) and a *monotonicity ratio* (fraction of steps moving in the expected direction).

2.3 The reliability gap and ReCalX calibration

Heavily masked images lie far from the training distribution, so f 's softmax confidence becomes unreliable — typically over-confident — and the causal curve is biased and noisy. ReCalX (2025) formalise this link between *uncertainty calibration* and perturbation-based explanation quality and propose *ReCalX*: recalibrate the model for the specific perturbations used, without altering its predictions. We adopt a binned temperature-scaling realisation. The perturbation

level $\ell \in [0, 1]$ is split into $B=10$ bins; for each bin b we collect raw logits over a calibration set and fit a scalar temperature T_b by minimising cross-entropy with L-BFGS-B (bounds $[0.05, 5]$). At evaluation, the wrapper `ReCalXModel` selects the bin for the current step and returns z/T_b (softmax applied afterwards). Because $\arg \max_c z_c$ is invariant to dividing by $T_b > 0$, calibrated and raw models make identical predictions; only the confidence is corrected.

2.4 Significance bucketing

The classic schedule removes a fixed top- $n\%$ (e.g. 1%) per step. This is fragile for sparse/skewed maps: when most pixels share the same (near-zero) value, value-thresholding collapses them into one step. *Significance bucketing* instead sorts pixels by *rank* and removes N equal-count groups, one per step. Being rank-based it is scale-invariant (so IG, Saliency and RISE are treated identically) and its N steps line up with the calibration bins. We use $N=25$ buckets for evaluation and contrast against fixed-step 1% deletion.

2.5 Datasets, models, and the ViT bonus

We use an ImageNet validation subset (4,358 images available; 20 sampled per run) with standard preprocessing (resize 256, center-crop 224, ImageNet normalisation). Models are torchvision ImageNet-pretrained ResNet50, ResNet101 (He et al., 2016), ViT-B/16 and ViT-B/32 (Dosovitskiy et al., 2021); ResNet50 and ViT-B/16 agree on the top-1 class for 19/20 sampled images. As a theoretical bonus we surveyed three attribution methods designed specifically for Vision Transformers: (i) **Attention Rollout** (Abnar and Zuidema, 2020), which multiplies (residual-augmented) attention matrices across layers to trace token-to-patch influence; (ii) **Transformer Attribution / LRP relevance propagation** (Chefer et al., 2021a), a class-specific rule propagating relevance through attention and skip connections; and (iii) **Generic Attention-model Explainability** (Chefer et al., 2021b), which combines gradient-weighted attention across layers and generalises to multi-modal transformers. These are natural drop-in explainers for the ViT branch of our pipeline.

3. Experiments

All runs use $N=20$ images; calibration temperatures are fit jointly (batch) unless stated. ECE uses 10 equal-width bins; TACE uses 15 adaptive bins with threshold 0.01. Significance is assessed with the paired Wilcoxon signed-rank test ($\alpha=0.05$).

3.1 Phase 2 – Pre-calibration baseline

Table 1 reports per-method deletion AUC (lower = sharper confidence drop = better localisation) on ResNet50. Gradient methods (IG, GradientSHAP, Saliency) achieve the lowest deletion AUC, while LIME is highest. These raw numbers, however, are confounded by the reliability gap (next subsection): the curves they integrate are built on miscalibrated confidence.

3.2 Phase 3 – Calibrated causal analysis

The gap. On ResNet50 with RISE + bucket deletion, raw ECE/TACE rise sharply with perturbation, confirming OOD over-confidence. The fitted per-bin temperatures increase monotonically with perturbation (Table 2): heavier masking requires more softening, exactly the signature ReCalX predicts.

Table 1: Phase 2 baseline on ResNet50 (RISE-style deletion, $N=20$): mean deletion AUC and raw ECE per attribution method. Lower deletion AUC is better.

Method	Deletion AUC ↓	Raw ECE
Integrated Gradients	0.068 ± 0.065	0.162
GradientSHAP	0.080 ± 0.081	0.186
Saliency	0.111 ± 0.075	0.211
RISE	0.178 ± 0.113	0.281
Feature Ablation	0.205 ± 0.135	0.390
LIME	0.380 ± 0.136	0.422

Table 2: Batch-fitted per-bin temperatures (ResNet50, RISE). $T > 1$ softens over-confident logits; the monotone rise tracks the increasing severity of perturbation.

Bin (level)	0-.1	.1-.2	.2-.3	.3-.4	.4-.5	.5-.6	.6-.7	.7-.8	.8-.9	.9-1
T_b	0.24	0.52	0.57	0.55	0.60	0.67	0.70	0.79	0.96	1.37

Calibration reduces miscalibration. On ResNet50 (RISE, bucket-deletion, $N=20$) recalibration roughly halves both errors: ECE $0.282 \rightarrow 0.148$ and TACE $0.350 \rightarrow 0.160$; 15/20 images improve (mean $\Delta\text{ECE} +0.134$). The effect is largest where raw miscalibration is largest (Table 3): LIME ($\Delta\text{ECE} -0.328$), Feature Ablation (-0.186) and RISE (-0.134), and near-neutral for the already-sharp gradient maps (IG $+0.003$, GradientSHAP -0.010).

Calibration improves AUC and monotonicity. A Wilcoxon test (Table 4) shows deletion AUC increases significantly after calibration for *every* method ($p < 0.001$). Curve monotonicity (Spearman r , more negative = better for deletion) improves significantly for the noisy perturbation-based maps — RISE $-0.82 \rightarrow -0.94$ ($p=0.001$), LIME $-0.59 \rightarrow -0.94$ ($p < 0.001$), Feature Ablation $-0.48 \rightarrow -0.81$ ($p=0.006$) — while the gradient maps were already near-monotone and change little. This directly answers RQ1: recalibration makes the causal signal both larger and steadier, most where it was least reliable.

Across architectures. Calibration helps all four backbones (Table 5), with the largest AUC gain on ResNet50 and consistently more-negative Spearman after calibration. ViTs are better calibrated to begin with (raw ECE 0.20 vs 0.28 for ResNet50) and already produce the most monotone raw curves, so their headroom is smaller — but calibration still improves every metric.

3.3 Phase 3 – Significance bucketing vs fixed-step

Comparing the two perturbation schedules on ResNet50 + RISE (Table 6), significance bucketing produces more monotone curves than fixed-step 1% deletion, both raw and calibrated. The differences are statistically significant: Wilcoxon $p=0.0002$ for Spearman r and $p=0.0009$ for the monotonicity ratio. Calibration ECE is comparable between the two (0.128 for TopN vs 0.148 for Bucket), so bucketing buys stability without hurting calibration. This answers RQ2: rank-based bucketing is the more sensitive and robust schedule, and (being scale-invariant) it is the safer default for sparse attribution maps.

4. Conclusions

The insertion/deletion “reliability gap” is, to a large extent, a calibration artefact: under explainability-specific perturbations the classifier is pushed OOD and its confidence — the very quantity the AUC integrates — becomes untrustworthy. Perturbation-aware temperature scaling

Table 3: Per-method calibration effect on ResNet50 (ECE before/after, $N=20$). Negative ΔECE = improvement.

Method	ECE raw	ECE cal	ΔECE
LIME	0.422	0.094	-0.328
Feature Ablation	0.390	0.205	-0.186
RISE	0.281	0.148	-0.134
Saliency	0.211	0.168	-0.043
GradientSHAP	0.186	0.176	-0.010
Integrated Gradients	0.162	0.165	+0.003

Table 4: Effect of calibration on ResNet50 (Wilcoxon signed-rank, $N=20$). Δ values are medians; * marks $p < 0.05$.

Method	ΔAUC	$p(\text{AUC})$	$\Delta\text{Spearman}$	$p(\text{Spear})$
RISE	+0.250	0.000*	-0.064	0.001*
LIME	+0.340	0.000*	-0.309	0.000*
Feature Ablation	+0.309	0.000*	-0.254	0.006*
Saliency	+0.178	0.000*	-0.016	0.812
Integrated Grad.	+0.082	0.000*	+0.003	0.317
GradientSHAP	+0.099	0.000*	+0.012	0.059

(ReCalX) addresses this cheaply and post-hoc: it leaves predictions untouched, roughly halves ECE/TACE, significantly increases deletion AUC, and makes the causal curves significantly more monotone, with the biggest gains on the perturbation-based attributions and on CNNs. Rank-based significance bucketing further stabilises the test and is robust to sparse maps, significantly outperforming fixed-step removal in monotonicity. **Take-away:** calibrating the model *for the perturbation it is about to see*, and removing pixels by rank rather than by a fixed percentage, together turn insertion/deletion into a markedly more trustworthy XAI metric.

Limitations and future work. Evaluations use $N=20$ images, so effect sizes should be read as indicative; the masking baseline (black/blur) is itself OOD, motivating the planned inpainting-based generator (currently a stub); and single-image calibration overfits (temperatures hit the search bounds), so batch fitting is preferred. Promising extensions include superpixel/patch-based deletion aligned with ViT tokenisation and FID/OTDD distances to quantify how far each perturbation pushes inputs off-manifold.

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Table 5: Architecture comparison (RISE, bucket-deletion, $N=20$): deletion AUC and Spearman r before/after calibration.

Architecture	AUC raw	AUC cal	Spear raw	Spear cal
ResNet50	0.178	0.401	-0.816	-0.942
ResNet101	0.275	0.344	-0.833	-0.903
ViT-B/16	0.207	0.244	-0.924	-0.962
ViT-B/32	0.210	0.244	-0.901	-0.936

Table 6: Significance bucketing (25 buckets) vs fixed-step TopN (1%) on ResNet50 + RISE. Spearman r closer to -1 and higher monotonicity are better.

Schedule	Spear raw	Spear cal	Mono raw	Mono cal
TopN (1%)	-0.722	-0.908	0.557	0.583
Bucket (25)	-0.816	-0.942	0.632	0.686
Wilcoxon Bucket vs TopN: Spearman $p=0.0002^*$; Monotonicity $p=0.0009^*$.				

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A. Appendix: Additional Results

A.1 Integrated-Gradients cross-architecture run. Repeating the architecture comparison with IG instead of RISE (Table 7) reproduces the same qualitative pattern: calibration raises deletion AUC and makes Spearman r more negative on all four backbones.

Table 7: Architecture comparison with Integrated Gradients (bucket-deletion, $N=20$).

Architecture	AUC raw	AUC cal	Spear raw	Spear cal
ResNet50	0.081	0.163	-0.900	-0.936
ResNet101	0.190	0.220	-0.879	-0.909
ViT-B/16	0.202	0.236	-0.909	-0.935
ViT-B/32	0.173	0.213	-0.954	-0.969

A.2 TopN vs Bucket calibration error. ECE/TACE before→after calibration: TopN(1%) ECE 0.278→0.128, TACE 0.303→0.132; Bucket(25) ECE 0.281→0.148, TACE 0.350→0.160. Both schedules calibrate comparably well.

A.3 Spearman/monotonicity per method (raw → cal), ResNet50 + RISE. Feature Ablation $-0.48 \rightarrow -0.81$, LIME $-0.59 \rightarrow -0.94$, RISE $-0.82 \rightarrow -0.94$, Saliency $-0.89 \rightarrow -0.79$,

GradientSHAP $-0.93 \rightarrow -0.79$, IG $-0.93 \rightarrow -0.81$. Calibration helps the noisy (perturbation-based) maps and is roughly neutral-to-slightly-negative for maps that were already highly monotone.

A.4 Normalisation sanity check. Raw attribution ranges differ wildly across methods (e.g. IG $[-1.01, 1.14]$, RISE $[0.18, 1.21]$, GradientSHAP $[-1.34, 1.52]$). Because the bucket generators use `argsort`, they are rank-invariant and unaffected by scale — one of the practical motivations for significance bucketing.